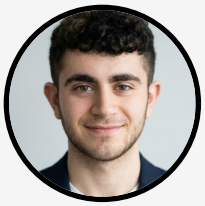
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Christian Santarelli

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ABOUT ME

I am a curious and driven individual with a strong passion for technology and innovation. I enjoy tackling complex problems with an analytical mindset and am always looking for opportunities to expand my skill set and contribute to ambitious projects. Strong attention to detail and the ability to adapt quickly to new environments define my professional profile.

EDUCATION & TRAINING

1 Sep 2023 - Current - TURIN, ITALY

BACHELOR'S DEGREE IN COMPUTER SCIENCE- UNIVERSITY OF TURIN

- **Algorithms, Data Structures & Computational Complexity**: Asymptotic algorithm design and analysis, advanced data structures (hash tables, trees, graphs, Union-Find disjoint sets), dynamic programming, string-matching algorithms (Rabin-Karp), and low-level development in C.
- **Operating Systems, Concurrency & Systems Programming**: Computer architecture, process and memory management, concurrent synchronization, inter-process communication (IPC), Linux systems programming (C, Rust, x86_64 Assembly), and telemetry daemons for HPC/AI infrastructures.
- **Formal Languages, Automata & Compilers**: Computability theory, finite state automata (DFA/NFA), regular expressions, lexical analysis, top-down LL(1) recursive descent parsing, Syntax-Directed Translation (SDT), and JVM bytecode generation (Jasmin).
- **Software Engineering & Object-Oriented Architectures**: Requirements engineering, UML modeling (Use Cases, Domain Models, SSD, DSD, DCD), GRASP principles and GoF Design Patterns (State, Strategy, Observer, Factory, Singleton), data persistence, and automated testing (JUnit).
- **Database Systems & Information Engineering**: Conceptual design (E-R modeling and business rules), logical design (volume, operation, and redundancy analysis, normalization), SQL development (DDL, DML, triggers, integrity constraints), and query optimization on relational DBMSs (PostgreSQL, SQLite).
- **Web Technologies, Cloud-Native Backends & Observability**: Enterprise RESTful architectures with Java and Spring Boot, stateless JWT authentication and Spring Security, ORM (JPA/Hibernate), Docker containerization, performance profiling, and observability stacks (Prometheus, Grafana, Loki, Promtail).
- **Distributed Systems, Cryptography & Blockchain Protocols**: EVM smart contracts in Solidity (Hardhat, Factory-Instance and Pull-over-Push patterns, anti-reentrancy security, gas optimization) and Bitcoin Layer-1/Layer-2 protocols in Python (Taproot/Schnorr, Lightning Network BOLT, Eltoo and PTLC channels, Sphinx onion routing, Submarine Swaps, asynchronous P2P daemons).
- **Human-Computer Interaction (HCI), Frontend Engineering & Data Science**: Design of accessible and internationalized user interfaces (React, TypeScript, Next.js, Tailwind CSS), data engineering, and Exploratory Data Analysis in Python (data-cleaning pipelines, univariate/multivariate statistical analysis, time series, geospatial analysis, and anomaly detection).

Field(s) of study: Natural sciences, mathematics and statistics not further defined | **Level in EQF:** 6 | **National classification:** 6 | **Type of credits:** ECTS | **Number of credits:** 180 | **Thesis:** Comprehensive Datacenter Usage and Sensor Monitoring | **Website:** unito.it

WORK EXPERIENCE

1 AUG 2023 - 30 MAY 2024 - PIOSSASCO, ITALY

SOFTWARE DEVELOPER POWERLINE 2.0

- End-to-end design and full-stack development of a dynamic web platform for a fitness center (powerline-piossasco.it):
  - Conducted requirements analysis, system architecture design, relational database modeling, and GDPR-compliant sensitive data handling.
  - Developed the web application using **PHP, HTML5, CSS3 (Bulma), and JavaScript (jQuery)**, delivering a client portal (membership status and digital workout distribution) and an administrative staff dashboard (CMS for homepage assets, user registrations, subscriptions, and personalized training schedules).
  - Executed end-to-end functional and integration testing

UNIVERSITY OF TURIN - DEPARTMENT OF COMPUTER SCIENCE - TURIN, ITALY

TUTOR DIDATTICO DI LABORATORIO 3 OCT 2025 - 9 DEC 2025

- Provided hands-on laboratory tutoring and technical assistance for the *Algorithms and Data Structures* course (B.Sc. in Computer Science), working alongside the tenure-track professor to support a cohort of ~100 students.
- Supervised practical programming sessions focused on implementing, debugging, and optimizing fundamental data structures and algorithms.
- Conducted real-time code reviews, assisted with memory-management troubleshooting and edge-case handling, and guided students through empirical and asymptotic computational complexity analysis.

ACADEMIC TUTOR & ALGORITHMIC ASSESSMENT DESIGNER 11 APR 2026 - 23 SEP 2026

- Collaborated with the theory professor of the *Algorithms and Data Structures* course to design and develop automated grading exercises on **Moodle (CodeRunner plugin)** for the upcoming academic year.
- Conceived and implemented a 4–5 stage guided composite exercise requiring students to build an advanced variant of the **Rabin-Karp** algorithm (fixed-length multi-pattern string matching) from scratch.
- Engineered rigorous test suites and anti-brute-force validation mechanics (tracking verified positions to enforce rolling-hash time complexity constraints), overcoming CodeRunner's output-only evaluation limitations.
- Optimized visual rendering and code formatting across Moodle problem statements by managing HTML/CSS syntax-filter behaviors.

SKILLS

computer programming | interpret technical requirements | SQL | object-oriented modelling | Gestione autonoma della posta e-mail | c | c++ | c# | java | JavaScript | TypeScript | React | NextJS | Conteinerizzazione e gestione di pod (wildfly, apache, nginx) tramite Docker e Kubernetes | Python (computer programming) | Spring (Spring Boot Spring Data Spring Integration) | PHP | MySQL | Database non relazioni (MongoDB) | understand spoken English | engineering processes | interact verbally in English | html | css | sql | javascript | mysql | python | php | linux | Windows | xml | git | bootstrap | jquery | perform scientific research | engineering principles | debug software | web services | integrated development environment software | use an application-specific interface | use software libraries | analyse software specifications | ICT debugging tools | create flowchart diagram | develop software prototype | project management | manage engineering project | rust | Sviluppo di applicazioni enterprise ad alta concorrenza, throughput ed affidabilità | json | ajax | Risoluzione dei problemi | Tokio | metaprogrammazione | use software design patterns | use scripting programming | HPC | redfish | Protocolli industriali (modbus, mbus, profibus, profinet) | Pandora 9.0.0 | NVIDIA management library | Database Time Series: InfluxDB | Containerizzazione: Docker, Kubernetes | GitHub / GitLab | makefile | tools for software configuration management | cargo workspace

LANGUAGE SKILLS

Mother tongue(s): **ITALIAN**

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	C1	C1	B2	B2	B2

DRIVING LICENCE

Driving licence: AM | 19 Sep 2023 - 13 May 2034

Driving licence: B | 19 Sep 2023 - 13 May 2034

HOBBIES AND INTERESTS

Emerging Technologies I closely follow the evolution of digital innovations, dedicating part of my free time to studying new tools and software and experimenting hands-on whenever possible.

PROJECTS

1 APR 2026 - 20 SEP 2026

YATAL: Real-Time Telemetry and High-Performance Monitoring for HPC & AI Datacenters Bachelor's thesis project developed at the Department of Computer Science of the University of Turin for the HPC4AI (High-Performance Computing for Artificial Intelligence) supercomputing infrastructure, under the supervision of Prof. Robert René Maria Birke (in memoriam).

Engineering Goals & Challenges: Design and implementation of a Linux-native telemetry daemon (systemd) written in Rust, engineered to eliminate the probe effect and address protocol fragmentation across scientific computing and AI clusters.

Key Results & Software Architecture:

- **Hybrid Sync-Async Architecture**: Integrated a multi-worker Tokio asynchronous reactor with dedicated blocking threads (spawn_blocking) for hardware sensor polling, ensuring that blocking system calls never stall network export pipelines.
- **Temporal Correctness Sentinels**: Formulated and implemented algorithms for clock-drift compensation in Out-of-Band controllers (Redfish BMCs) and active staleness filtering on monitoring brokers (Pandora FMS), eliminating ghost writes and rejected data points on InfluxDB.
- **Zero-Allocation Memory Engineering**: Optimized the hot path using SmallVec, structural tag sharing (Arc), and lock-free thread-local recycling pools (LOCAL_POOL), achieving zero lock contention and zero heap allocations during steady-state sampling.
- **Comparative Allocator Study & io_uring**: Conducted empirical benchmarks across jemalloc, mimalloc, and TCMalloc, identifying memory-pinning risks in long-running daemons and implementing asynchronous memory release (madvise), alongside high-performance local logging via Linux kernel io_uring with graceful fallback.
- **Operational Metrics**: Contained memory footprint at ~53 MB RSS in production with 9 active collectors across 14 threads, serialization throughput exceeding 1.35 million points/sec, and 90+ automated tests with GitLab CI/CD coverage.

1 SEP 2025 - 26 JAN 2026

GreenHabits Full-stack web platform built with an enterprise architecture. Implemented with a Java Spring Boot backend, a React/Vite (TypeScript) frontend, and a PostgreSQL database. The infrastructure is fully containerized with Docker and features an advanced observability and monitoring stack powered by Grafana and Prometheus.

Link https://github.com/Seintian/unito-web-technologies-25-26

1 SEP 2025 - 10 FEB 2026

School Outreach Portal High-fidelity, fully functional web application designed for the Department of Computer Science at the University of Turin. It serves as a dedicated portal for schools (students, teachers, and principals) to explore orientation activities, laboratories, and academic programs. Developed for the Human-Computer Interaction course, the project aggregates various Public Engagement initiatives (such as Social4School and AIRLab) into a single accessible, user-centered platform.
Links https://seintian.github.io/unito-human-computer-interaction-25-26 |

https://github.com/Seintian/unito-human-computer-interaction-25-26

28 JAN 2026 - CURRENT

WinHUD Lightweight, non-intrusive system performance overlay for Windows. It provides real-time monitoring of critical hardware metrics (CPU, GPU, RAM, Disk, Network) without cluttering the screen or stealing window focus from games. Designed around a "set it and forget it" philosophy, WinHUD automatically detects active gaming sessions, dynamically adapts text contrast to background luminance, and never obstructs gameplay thanks to a click-through "Ghost Mode".
Link https://github.com/Seintian/WinHUD

8 FEB 2026 - CURRENT

libnumakit High-performance C toolkit designed to bridge the gap between hardware topology discovery (e.g., hwloc) and application logic on modern multi-socket NUMA systems. It provides topology-aware memory allocators, low-latency synchronization primitives (MCS locks), and thread affinity policies. By strictly governing memory placement and eliminating cross-socket interconnect traffic during lock contention, it prevents critical performance bottlenecks such as false sharing and remote-allocation overhead.
Link https://github.com/Seintian/libnumakit

1 SEP 2025 - 12 FEB 2026

Formal Languages and Translators Complete, from-scratch implementation in Java of a compiler for a custom procedural programming language. The architecture follows a classic multi-pass design: a custom lexical analyzer tokenizes the source code, an LL(1) recursive descent parser constructs an Abstract Syntax Tree (AST), and a semantic analysis pass enforces type safety and scoping rules via Syntax-Directed Translation, culminating in executable JVM bytecode generation. The target language features C-like control flow (while, conditional branches) combined with prefix (Polish) notation for arithmetic and boolean expressions, supporting n-ary operators and statement blocks.
Links https://github.com/Seintian/unito-formal-languages-and-compilers-25-26 |

https://github.com/Seintian/unito-formal-languages-and-compilers-25-26/blob/master/GRAMMAR.md

1 NOV 2024 - 31 JUL 2025

POSIX Operating Systems Simulation Comprehensive multi-process simulation framework developed for the Operating Systems course (A.Y. 2024–2025). The simulation models a post-office service ecosystem with interacting concurrent actors: Director, Workers, Users, and Ticket Issuers. Built on a modular architecture in C, the system orchestrates bidirectional communication and synchronization across processes using advanced System V IPC primitives. A companion terminal-based interactive UI application allows users to configure simulation parameters and inspect live logs. For fault tolerance, the system autonomously handles crash scenarios by generating post-mortem crash reports with CPU register dumps on both x86-64 and aarch64 architectures.
Link https://github.com/Seintian/unito-operative-systems-24-25

1 SEP 2024 - 27 JAN 2025

Algorithms & Data Structures Library Comprehensive suite of C libraries implementing fundamental data structures and algorithms from scratch for the Algorithms and Data Structures course, engineered to maximize time and space efficiency. Developed using Test-Driven Development (TDD) with the Unity C testing framework, ensuring reliability through intensive unit testing across every module. Modules span sorting algorithms, dynamic-programming Edit Distance computation for text correction, generic Hash Tables, and sparse Graphs with Breadth-First Search (BFS) traversal.
Link https://github.com/Seintian/unito-algorithms-and-data-structures-24-25

1 SEP 2025 - 10 FEB 2026

Data Analysis - Anime Data Storytelling End-to-end data processing and exploratory analysis pipeline investigating a massive dataset of over 130 million data points covering the anime industry (titles, characters, voice actors, and user interactions). Developed in Python within JupyterLab using the uv package manager for deterministic, high-speed dependency resolution. Beyond statistical aggregation, the project emphasizes visual Data Storytelling—uncovering structural patterns, historical trends, and statistical anomalies to deliver actionable insights.

Link https://github.com/Seintian/unito-hci-data-analysis-25-26

CERTIFICATIONS

8 MAY 2026 NVIDIA DLI

Data Parallelism: How to Train Deep Learning Models on Multiple GPUs

Online - Online – 8-hour instructor-led workshop organized by the NVIDIA Deep Learning Institute (DLI). Covered data-parallel training of deep learning models across multiple GPUs using PyTorch Distributed Data Parallel (DDP) and NCCL. Key topics included stochastic gradient descent and the impact of batch size on training time and convergence accuracy, refactoring single-GPU training pipelines to multi-GPU with DDP, and algorithmic techniques to preserve model accuracy at scale. Passed a hands-on coding assessment and earned an NVIDIA DLI Certificate of Competency.

Link https://learn.nvidia.com/certificates?id=DF4ratM2TJiNiUKrURoGbQ

28 JUL 2022 Cambridge Assessment English

First Certificate in English

Presential - In-person – Cambridge English: First (FCE) certification, CEFR Level B2. Overall score: 177 (Grade B). Assessed skills: Reading (178), Use of English (185), Writing (173), Listening (177), Speaking (172). Exam taken on May 25, 2022, in Turin.

27 JAN 2021 Synergie Italia (tramite Forma.Temp)

General Workplace Safety Training Course – Legislative Decree 81/08 – State-Regions Agreement 21/12/11

Presential - In-person – 4-hour theoretical training course on occupational health and safety pursuant to Italian Legislative Decree 81/08. Covered regulatory frameworks, risk assessment, occupational hygiene, rights and duties of employers and employees, Personal Protective Equipment (PPE) usage, and corporate safety organization.

24 MAR 2022 Liceo Scientifico Statale "Maria Curie" – Pinerolo

Student-Worker Occupational Safety Training – Public Education – Specific Risks (Art. 37, Legislative Decree 81/08)

Presential - In-person – 8-hour training course on specific occupational hazards pursuant to Italian Legislative Decree 81/08. Covered risk assessment, facility safety features, worker obligations, biological, chemical, fire, and VDT (video display terminal) hazards, work-related stress, PPE, and emergency evacuation procedures.